2021 International Conference on Computer Information Science and Artificial Intelligence (CISAI) | 978-1-6654-0692-5/21/$31.00 ©2021 IEEE | DOI: 10.1109/CISAI54367.2021.00172

Application of artificial intelligence in clothing intelligence manufacturing

Yalin Dong*
Nanchang Institute of Science & Technology, Nanchang, Jiangxi, 330108, China
*Corresponding author's e-mail: 739960103@qq.com

***Abstract*—At present, the rapid development of science and technology in China not only provides solid scientific and technological support for the development of various intelligent technology products, but also opens up a new road for intelligent clothing manufacturing. Many aspects of artificial intelligence, such as industrial chain and computer application, can make a contribution to clothing intelligent manufacturing. This paper first introduces the relevant information of artificial intelligence, then explains the specific application direction of artificial intelligence in clothing intelligence manufacturing, and finally looks forward to its application prospects.**

Keywords-Artificial intelligence; clothing intelligence manufacturing; application

I. INTRODUCTION

This year, China put forward the goal of "Made in China 2025". The specific content is to clarify the tasks and priorities of a manufacturing country through planning a series of industrial layout. The final goal is to achieve the transformation from an industrial country to an industrial power. Under the guidance of the national policy, many traditional clothing industry takes the internal reform step by step with the goal of intelligence and digitalization step by step. To this end, a large number of scholars discuss around the theme of clothing intelligence, in order to use the advantages of artificial intelligence into clothing intelligent manufacturing. In fact, in 2017, the State Council issued a plan to support the development of artificial intelligence, which made the requirements for the scale of the artificial intelligence industry every five years. However, the theoretical requirements alone are not enough. The relevant departments of the state released the "5 G Commercial Application Plan" in 2019 to implement the project of artificial intelligence on the basis of theory. We believe that with the strong support of the country, artificial intelligence will flourish in the land of China and bring science and technology to major Chinese enterprises.

II. AI AND INTELLIGENCE MANUFACTURING-RELATED INFORMATION

Since China released the strategic goal of "Made in China 2025" in 2015, the whole country has formed a strong intelligent manufacturing force. Because of this strategy, China will transform from a manufacturing country into a manufacturing power within 30 years. Among them, when the Ministry of Industry and Information Technology interpreted the specific content of the strategy for the public, it was put forward that in order to complete the industrial transformation within 30 years, China must use the strength of the whole country to work together. At the same time, intelligent machine production methods should also be used in China's manufacturing sector. The reason for this approach is that intelligent manufacturing is based on big data and computer systems, and only after the factory is it on a large scale can their industry move in the direction of intelligence. In addition, intelligent manufacturing has a series of conventional machines. Specifically, AI has the potential in five aspects. The first is the active perception, after the programmer has assembled the language and body action system for the artificial intelligence, the artificial intelligence can actively identify the person's state at that time. In addition, it will also intelligently analyze the meaning of people doing actions in this scene, and then extract the information in the knowledge base, and quickly respond to people, so as to achieve a good effect of human-machine interaction. The second is active learning. Living in a ever-changing society, every minute produces a lot of information. So the information stored in AI is definitely not constant, On the contrary, it can actively learn part of the knowledge according to people's actual needs. The third is conscious decision-making. Generally speaking, the occurrence of accidents has a certain contingency, so people need to make quick decisions in a short time to reduce unnecessary losses. And AI can provide people with scientific and reasonable decision-making opinions out of emergency. Because artificial intelligence is essentially the embodiment of countless running programs, so it does not have the human emotion, and there is no panic mood in danger. So AI can stand in a completely objective and rational perspective, make a suggestion to people out of panic. For example, when a fire occurs inside a factory due to aging circuits, artificial intelligence quickly activates the alarm device and makes harsh noises to remind people to escape the fire quickly in a safe manner. The third is automatic enforceability. The characteristics of this aspect is that AI can accomplish tasks under pre-programmed people without external factors. Automation also means that AI already has the ability to deal with periods, and one can deliver complex things to it. This way, AI can deal with things quickly and provide a relatively quiet space. Finally, it is automatic adaptation. This is manifested in that AI can adapt to changing environments and needs and complete its work according to the specific requirements at the time. Because the social environment is constantly changing, AI will not only do fixed jobs in one job. So the researchers have designed a large number of environmental perception systems for AI to help them better integrate into different environments[1].

978-1-6654-0692-5/21/$31.00 ©2021 IEEE
DOI 10.1109/CISAI54367.2021.00172

Authorized licensed use limited to: Pontificia Universidad Javeriana. Downloaded on July 27,2023 at 00:26:51 UTC from IEEE Xplore. Restrictions apply.

III. Specific application of artificial intelligence in the clothing manufacturing industry

A. *The relationship between AI and clothing manufacturing*

In the current rapid development of science and technology, both the handicraft industry and the manufacturing industry are inevitably under the impact of science and technology. Clothing design is an industry with a long history. In this industry, not only designers should change the traditional concept, and look at clothing design from a new idea and perspective. And clothing processing manufacturers also need to use artificial intelligence in the era of intelligence, and to provide the efficiency and overall level of clothing production. Artificial intelligence has undoubtedly brought a huge impact on the overall clothing manufacturing, and the latter in a new era background, must not be complacent. On the contrary, it should look at the road of clothing intelligence with full enthusiasm. At present, some developed regions in China are trying to apply artificial intelligence to the field of intelligent design. This means that the designer's dominance as a person is declining in the field of clothing intelligent manufacturing, while computers as a brand new thing, its impact on clothing design is deepening. But no matter where the process of history develops, people are always the subject full of creativity and vitality, and he can have a considerable impact on any industry. To sum up, artificial intelligence has played a great deal of role in clothing design. Next, I will specifically discuss the application of artificial intelligence in the intelligent manufacturing of clothing enterprises.

Specific application can be divided into three aspects. First, implant artificial intelligence as an element into the product, so that the original single product radiates the power of technology. After adding scientific elements to the product, it not only indirectly adds added value to the product, but also provides more consumption choices for people's lives. Any manufacturing product, under the processing of artificial intelligence, can become a new generation of intelligence products. But what we should see is that this behavior of implanted elements is only a one-sided combination of products and technology, and does not see the characteristics of the product itself. Therefore, this behavior is also a lower combination of artificial intelligence and manufacturing. Second, the after-sales service link of artificial intelligence elements, different from the first way, which provides consumers with a good service environment. Because a product is created by layer of processing, especially the product with the nature of technology, manufacturers should not only ensure the quality of the product to consumers, but also pay close attention to the use of customers and after-sales service[2]. A commodity, if it does not achieve excellent quality and excellent after-sales level is unified, it can undoubtedly not win the trust and love of consumers. It is very important for merchants to provide high-quality service to customers, but at present, there is a contradiction between consumers and manufacturers, that is, the number of consumers is too large, and manufacturers do not pay attention to after-sales service. Therefore, in the context of the industry does not pay attention to after-sales service, manufacturers should find a way, or use advanced technology to make up for the gap in consumers. At the moment, artificial intelligence can use the form of service to reduce the after-sales pressure of businesses. Because of the convenience and intelligence characteristics of artificial intelligence, it can quickly find out problems from customer feedback, and give the right solution. At the same time, manufacturers can also enter common interpersonal gestures and passwords inside the artificial intelligence to achieve a more realistic effect of serving people. Finally, integrate artificial intelligence elements into production, making manufacturing into intelligent manufacturing with technological nature. The last requirement of this aspect is to view the integration of artificial intelligence and manufacturing industry from a more macro perspective, and also puts forward a higher level of requirements for the major industries. In the early 21st century, a certain American authority issued a report called the "Artificial Intelligence Development Report in the Early 21st Century". The report condenses the efforts and achievements of hundreds of people, and therefore is highly valuable, while it presents the significant impact of AI on enterprises, as shown in the figure below. From the following table, AI technology has been widely used in various fields of manufacturing, and has had a significant impact on them. At the same time, with the maturity of artificial intelligence, it will have increasing influence in enterprises. Of course, AI technology is also becoming increasingly involved in various projects.

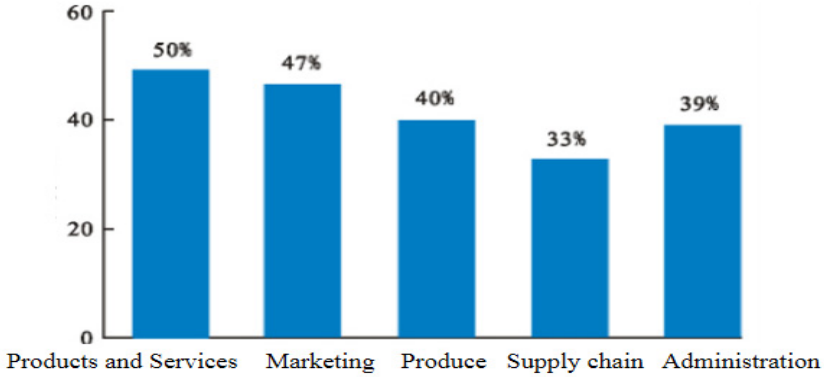

Figure 1. *AI significantly affects the business.*

Authorized licensed use limited to: Pontificia Universidad Javeriana. Downloaded on July 27,2023 at 00:26:51 UTC from IEEE Xplore. Restrictions apply.

B. *The intelligent application of artificial intelligence in clothing and accessories.*

On the basis of the current development of science and technology, if designers want to achieve the intelligent clothing in clothing, they can only rely on big data technology and clothing fabric technology. The integration of these two technologies can make the product have the intelligent attribute. The vast majority of intelligent clothing in the market also make use of these two science and technology in the production process. Producers help them develop smart products by hiring a group of people skilled in these technology. Of course, many producers have chosen to produce intelligent products under the strong market demand. For example, the smart slimming suit launched by an enterprise in Shenzhen at the beginning of this year is a vivid case of the combination of science and technology and clothing. This smart suit not only makes innovations in cutting fabrics, using the elastic fabric conducive to human movement, but also features smart products such as sensors on the clothes, in order to timely detect the wearer's actual feelings and feedback. At the same time, the dress uses a large number of muscle simulators, and then moves the body into the central nerve of the brain to stimulate the rhythmic movement of the muscles. Because the slimming suit will successfully complete the above processes, so it can help the wearer to exercise, in order to achieve a good effect of weight loss. The slimming has created huge consumption, and the company launches another smart bra while the iron is hot. The bra is different from the clothing available in that it monitors the local state of the wearer to keep the user in a healthy and stable range. The specific workflow of this bra is that it can establish a small temperature field within the wearing range and monitor the temperature data in real time. By comparing the data obtained again with the normal data, if the charts presented by the two data are relatively consistent, it proves the user's health, and vice versa, there are certain problems. In addition, intelligent sports shoes and socks can also protect users when exercising. Generally speaking, people in exercise are prone to respiratory disorders, breathing disorders and arrhythmia and other phenomena, if not found in time, people may have life in danger. Therefore, intelligent shoes and socks are designed to provide a health level guarantee for the human body in the process of exercise. Through special production materials, people can use conduction fiber to transmit sports data to the connected mobile phone, where users can understand the speed and frequency of their running, as an important reference information for the next stage of training[3].

C. *Using artificial intelligence for intelligent clothing design.*

In people's traditional cognition, clothing design only requires designers to draw specific design drawings and then over to the factory for manufacturing. This assembly-line production mode can both save labor and material resources, but also bring a lot of choice space for consumers. But people's current living standards are constantly improving, corresponding to a creative clothing design. Clothing, as a way to show people's aesthetic appreciation, has been divorced from the original people's demand for clothing is just to cover the body, and it has become more and more in pursuit of beauty. Therefore, designers should consider more factors when designing clothes, rather than simply tailoring them. In fact, artificial intelligence can play a considerable role in the design of smart clothing. In the early 21st century, MIT made beneficial attempts at artificial intelligence in clothing intelligence manufacturing. The specific attempt process is that the college uses the network system in artificial intelligence technology to automatically generate mannequins, and the designers design clothes according to the specific mannequins. This design method breaks through the past way that designers design with the body data provided by users, and using the technological means of building the model, it can perfectly simulate the form of users putting on the design clothes. While users can feel the quality or bad of clothing themselves, it also gives relevant designers the opportunity to correct mistakes, so that the clothing is presented in front of users in an almost perfect state. In fact, it is not common for MIT to use artificial intelligence technology in clothing design. In the current era of the rapid development of information, if the clothing design still blindly pursues the old design ideas, it is undoubtedly not conducive to the improvement of the overall level of the clothing industry. In the intelligent clothing designed by MIT, electronic wafers and multiple neural networks are used, striving to accurately respond to the current physical situation through electronic judgment. In addition, the design idea of intelligent clothing is not that the designer comes out of empty imagination, but that the designer obtains thousands of design drawings related to the theme of the work through the computer, and then takes its essence to design the style of intelligent clothing. The researchers believe that both smart and ordinary clothes are essentially serving people themselves, so the designer specially designed the clothing fabric as a soft and breathable special material. So when users put on smart clothes, they do not feel discomfort, on the contrary, they will feel more relaxed and carefree. When artificial intelligence has the above advantages in the field of clothing design, it will inevitably produce certain disadvantages. For example, intelligence means that the machine itself dominates its own behavior, while it obtains information corresponding to make decisions, which means that the machine collects the body part of its health information, such as heart rate level. A large number of users do not look at the user agreements before the use of smart clothing, which allows users to ignore the provisions on information sharing in the regulations, so even the manufacturers may be without full knowledge. Personal information is undoubtedly very important to users themselves, because in cyberspace, users get the right of way with one personal information after another. Once the user's information is leaked, in the era of intelligent intelligence, it is difficult to find out which intelligent product leads to this situation. To sum up, the use of any science and technology has its double-sided, and the producer of the intelligent product should give full consideration to the dual impact of the product on the user. When they use intelligent products, users should also retain a rational reason and thinking, and fully enjoy the convenience of intelligent products to life without revealing their own information[4].

Authorized licensed use limited to: Pontificia Universidad Javeriana. Downloaded on July 27,2023 at 00:26:51 UTC from IEEE Xplore. Restrictions apply.

D. *Artificial intelligence brings large-scale production mode to clothing intelligent manufacturing.*

Scale means that products can be mass produced, while ensuring the quality of each product. In a modern life, any production activity needs to improve efficiency to meet the growing consumer living needs and the profits of producers. The large-scale production mode was proposed in 1960 and continuously improved through a series of reforms of the production mode. It was not until the early 21st century that the large-scale production mode became the main mode of production of the factory. But we should also see that scale also means that products are completely produced through the assembly line and lack human wisdom input. Therefore, while undergoing mass production, it is also easy to ignore users' personalized clothing needs. In today's society, although there are a large number of psychological seeking psychology, there are some users seek differences, that is, different from others. Now a large number of single product market has tended to be saturated, and if manufacturers want to open up a new clothing market, they have to produce clothing products for this part that meet their needs from a personalized perspective of the group. Based on the above phenomenon, the researchers decided to use artificial intelligence, such a form of technology, to create production lines independent of intelligent clothing. The production line contains a variety of aspects, such as product scale, intelligent overall production and raw material supply automation. The intelligent production refers to the establishment of several small sites in the production line, each site is in charge of the manufacturing of a part, there is no connection between the site and the site. Such a production mode is convenient for personnel to each site regular patrol, if the product problem, personnel can also find the right site according to the process. After each parts of the smart clothing are processed, the corresponding fabric will be transmitted by a special smart robot. In the intelligent product line, the vast majority of processing is completed by the production line, which not only reduces the repetitive workload, but also greatly improves the production efficiency of products. In recent years, the traditional clothing department has been continuously carried out internal reform to meet the needs of artificial intelligence production. Now this reform has initially shown its unique advantages, so more and more industries have followed this way. This means that the clothing intelligent manufacturing industry is no longer a dominant scene, but a number of enterprises participate in, and gradually formed the development trend of internal competition and cooperation. We believe that the benign competition between enterprises will bring about the continuous improvement in product quality. The relevant departments of the state are also keenly aware of this phenomenon, so they have also introduced a series of policies to standardize enterprise competition in legal form, bring a clear and orderly environment for the market, and provide consumers with enough reassuring guarantee[5].

E. *Artificial intelligence is sold in smart clothing stores.*

Using AI in offline stores is a brand new attempt, because in the past stores, there were generally shopping guides to provide advice for customers to buy clothes. Artificial intelligence brings a new concept, that is, smart stores. Smart stores and traditional stores are different is that the former uses big data calculation and other ways to understand customers' preferences in advance, and then the robot will guide customers to find the clothes suitable for them. This way of using big data not only helps merchants improve sales, but also helps customers buy clothes at the wrong peak, It can be said to kill many birds with one stone. At present, a clothing store in Denmark has adopted this sales plan, which has also made innovations based on artificial intelligence services by using the function of facial recognition in front of the store to let the waiters know their names before they arrive. When the customer steps into the store, the waiter will greet him kindly. This method can let customers also feel a warmth of human communication in an intelligent store, so customers are more willing to come to the store, and lead friends to visit together.

IV. Application prospect outlook

At present, AI technology is booming and can be seen in every factory. Such a popularization of artificial intelligence is not only an affirmation of China's current science and technology level, but also a practical need of China's high-quality development. Therefore, this is the reason why the application of artificial intelligence in clothing intelligence manufacturing is here. Artificial intelligence technology is seizing the space of clothing manufacturing step by step, which are widely distributed in all aspects of clothing product design, mass processing, warehousing and logistics and even offline sales. Therefore, the comprehensive implementation of clothing intelligence manufacturing will be constantly close to the reality along with the development of artificial intelligence. Currently, AI is extremely accurate in all aspects, with even over 90 percent accuracy in image recognition and language processing. This also means that in the future sales process of intelligent clothing, AI will undertake the tasks of customer communication and communication, and positions such as shopping guide will continue to reduce until disappear. Of course, the essence of clothing intelligent manufacturing is to constantly add scientific and technological elements in clothing, so that artificial intelligence becomes a bridge of communication between producers and consumers. At the same time, the relevant researchers are also trying to use artificial intelligence to provide customers with high-quality services, to build a modular intelligence factory. Artificial intelligence is moving forward rapidly, and not in the 1920 s and 1930 s, AI will occupy its important position in various industries. It is predicted that in 2025, more than half of the enterprises will apply AI in the production process[6].

V. Conclusion

With its unique superiority, artificial intelligence occupies half of the clothing intelligence manufacturing. But both producers and consumers should be cautious while using smart technology, so that people can truly enjoy the benefits of technology.

Acknowledgement

1, Institute of Education Science of China Academy of Management Sciences: Research on the Relationship between Clothing Display Design and Clothing Retail Terminal

Authorized licensed use limited to: Pontificia Universidad Javeriana. Downloaded on July 27,2023 at 00:26:51 UTC from IEEE Xplore. Restrictions apply.

Marketing Strategy (Project No.: ZGYJKS2011346); 2. Jiangxi Art Science Planning Project: Application of Jiangxi Nuo Cultural Elements in Modern Clothing Design —— takes Nanfeng County as an Example (Project No.: YG2015087); —— in Traditional Tea Picking Opera takes Nanchang Tea Picking as an Example (Project No.: YG2014236).

REFERENCES

[1] Xia,M.(2019)The application status and prospect of intelligent manufacturing in the textile and garment industry.China textile.11:168-169.

[2] Yang,B.Q.ZhangJ,M.Yi,L.Y.etc.(2019)Research on the application of intelligent station terminal in the intelligent manufacturing industry of outdoor leisure products.Internet of Things technology.7:107-109.

[3] Liao,W.D.Ma,Y.Y.Liao,X.S.etc.(2019)Preliminary Discussion on Production Scheduling Optimization of Garment Intelligent Manufacturing System based on Big Data.Scientific and technological innovation.1:89-90.

[4] Zhang,H.(2019)Based on clothing intelligent manufacturing, three-dimensional cutting model digital correction research.Western leather.5:124-125.

[5] Wu,L.(20190Application of RFID technology in the intelligent manufacturing of the clothing industry.Journal of Xi'an University of Engineering.32:31-37.

[6] Yang,X.Y.Du,J.S.Ling,J.(2019)Factors affecting the maturity of clothing intelligent manufacturing capacity.Journal of Basic Science of Textile University.4:378-384.

Authorized licensed use limited to: Pontificia Universidad Javeriana. Downloaded on July 27,2023 at 00:26:51 UTC from IEEE Xplore. Restrictions apply.